

Project Design Phase-II

Data Flow Diagram & User Stories

Date	4 September 2026
Team ID	Servicenow
Project Name	Automated Network Request Management
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) is a visual representation of how data moves through the Automated Network Request Management system — how a request enters the system, how it is validated, approved, provisioned, and logged, and where data is stored along the way.

Level 0 – Context Diagram

Shows the system as a single process and its interactions with the three external entities: the Employee/Requester, the Approver/Manager, and the Network Admin (IT Team).

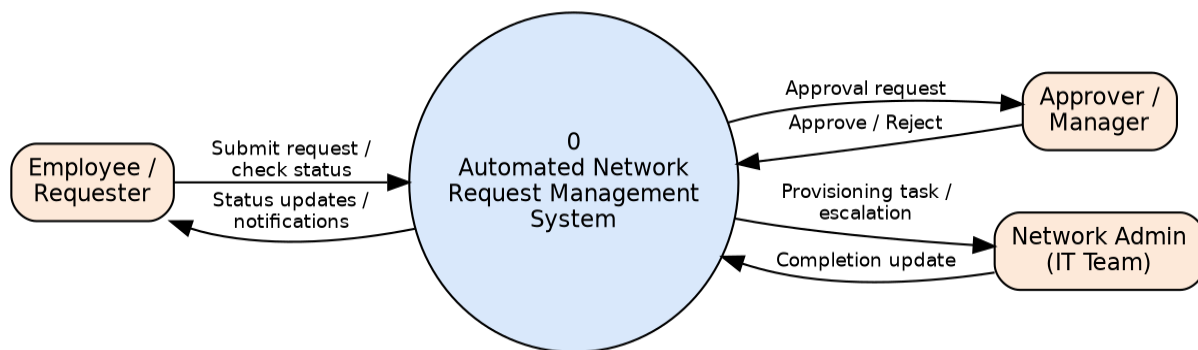


Figure 1: Level 0 Context DFD – Automated Network Request Management System

Level 1 – Detailed DFD

Decomposes the system into its core processes — authentication, request validation, approval routing, automated provisioning, notification, and logging/reporting — along with the data stores each process reads from or writes to.

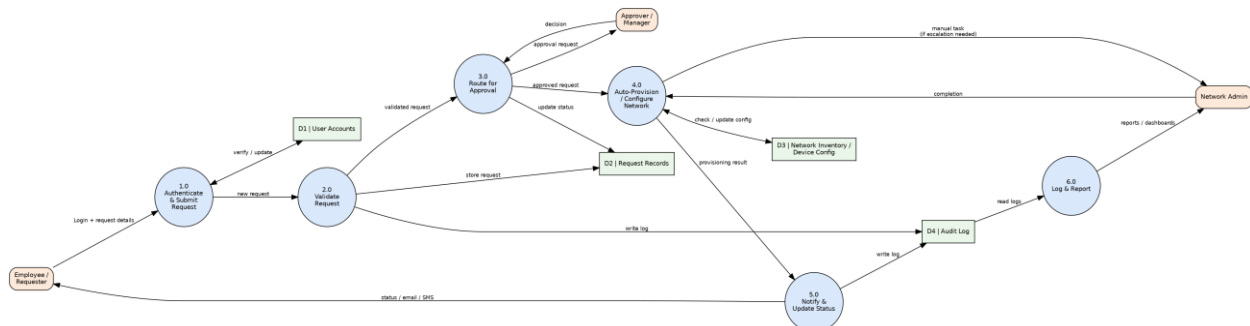


Figure 2: Level 1 DFD – Automated Network Request Management System

User Stories

The table below lists the user stories for the Automated Network Request Management product, grouped by user type and functional requirement (epic).

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee (Requester)	Registration & Login	USN-1	As an employee, I can register on the portal using my company email and employee ID so that I can raise network requests.	Employee can log in successfully with the registered credentials and lands on the dashboard.	High	Sprint-1
Employee (Requester)	Registration & Login	USN-2	As an employee, I can log in using corporate Single Sign-On (SSO) instead of a separate password.	User is redirected to the dashboard immediately after successful SSO authentication.	High	Sprint-1
Employee (Requester)	Request Submission	USN-3	As an employee, I can submit a network request (VPN access, Wi-Fi access, firewall rule, port opening) by selecting a request type and filling in the required details.	Request is saved with a unique ticket ID and initial status "Pending Approval".	High	Sprint-1
Employee (Requester)	Request Submission	USN-4	As an employee, I can attach supporting documents (e.g., manager authorization) to my request.	Uploaded file is stored securely and linked to the correct ticket ID.	Medium	Sprint-2
Employee (Requester)	Request Submission	USN-5	As an employee, I can save an incomplete request as a draft and submit it later.	Draft is retained in "My Requests" until the employee submits or deletes it.	Low	Sprint-3
Employee (Requester)	Dashboard & Tracking	USN-6	As an employee, I can track the real-time status of all my submitted requests on a single dashboard.	Dashboard shows current stage (Pending / Approved / Provisioning / Completed / Rejected) for every request.	High	Sprint-1
Approver (Manager)	Approval Workflow	USN-7	As an approver, I receive an automated notification as soon as a request needs my approval.	Email and in-app notification are sent within 1 minute of request submission.	High	Sprint-1
Approver (Manager)	Approval Workflow	USN-8	As an approver, I can view full request details and approve or reject the request with comments.	Approval/rejection decision and comments are recorded against the ticket and visible to the requester.	High	Sprint-1
Approver (Manager)	Approval Workflow	USN-9	As an approver, I can delegate my approval authority to another approver when I am out of office.	Delegated approver can view and act on pending requests during the configured delegation period.	Medium	Sprint-3
Approver (Manager)	Approval Workflow	USN-10	As the system, requests that are not actioned within the defined SLA are automatically escalated to the next approval level.	Unactioned request is escalated and both approvers are notified once the SLA timer expires.	Medium	Sprint-2
System (Automation Engine)	Automated Provisioning	USN-11	As the system, I can automatically provision the requested network access once it is approved, without manual admin intervention, for standard request types.	Access is provisioned within the defined SLA and ticket status auto-updates to "Completed".	High	Sprint-2
System (Automation Engine)	Automated Provisioning	USN-12	As the system, I can validate the requested configuration against network inventory and security policy rules before provisioning.	Requests that violate policy are automatically blocked and routed to a network admin for review.	High	Sprint-2
System (Automation Engine)	Automated Provisioning	USN-13	As the system, I can automatically revoke or expire access on the date specified in the request.	Access is removed on the expiry date and the requester and admin are notified.	Medium	Sprint-3
Network Admin (IT Team)	Exception Handling	USN-14	As a network admin, I can view and manually complete any request that fails automated provisioning.	Failed requests appear in an exception queue with the failure reason and can be actioned manually.	High	Sprint-2

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Network Admin (IT Team)	Exception Handling	USN-15	As a network admin, I can view live network inventory and device configuration status from a dashboard.	Dashboard reflects current device/port utilization sourced from the network inventory data store.	Medium	Sprint-2
Employee / Network Admin	Notifications	USN-16	As an employee, I receive an email/SMS notification whenever the status of my request changes.	Notification is sent within 1 minute of any status change and includes the ticket ID.	High	Sprint-1
Administrator	Reporting & Audit	USN-17	As an administrator, I can generate reports on request volume, SLA compliance, and average approval turnaround time.	Report can be filtered by date range and department, and exported to CSV/PDF.	Medium	Sprint-3
Administrator	Reporting & Audit	USN-18	As an administrator, I can view a complete audit log of every action taken on a request for compliance purposes.	Every create/approve/reject/provision/revoke action is logged with timestamp and actor.	High	Sprint-2